

01 The gap has not moved: dashboards show what moved, never why

73%

of BI implementations fail on the diagnostic gap — not on technology

~21%

of employees actively use the dashboards that cannot answer "why"

~70%

of analyst time spent investigating alerts that turn out to be legitimate

22%

decline in detection accuracy past 30+ alerts reviewed a day

Three reasons the gap persists

- **Detection is not explanation.** Tools flag that a number moved. The mechanism stays a manual investigation.
- **Evidence lives in two worlds.** Metrics sit in the warehouse; causes sit in support tickets, CRM notes, deploy changelogs and news.
- **Confidence is unpriced.** No tool states how sure it is, so a leader cannot separate a real finding from a fluent-sounding guess.

Verified market voice

"Dashboards are great at telling you what's happening, but the moment you ask why..."

ThoughtSpot user · G2

"...correlations that aren't actually meaningful without manual review."

Tellius user · G2

"Sales Ops deletes columns in Salesforce. Marketing changes the definition of a lead bucket."

r/BusinessIntelligence

THE PATH A LEADER LIVES TODAY

1 Dashboard flags a drop

No cause, no confidence, no next step.

2 Analyst pulls data

Four or more tools, manually reconciled.

Day 1

3 Cross-references

Tickets, CRM notes, changelogs, news.

Day 2

4 Writes a narrative

Plausible, unranked, no stated confidence.

5 Window has closed

The decision was already taken without it.

Every category in this market produces an explanation. None of them declines to produce one.

02 The obvious fix is the dangerous one

"The capability you want for root-cause analysis actively works against the humility you need."

A model that reasons

A 2026 fidelity audit found an LLM **inverted the risk direction on 3 of 4 factors** in a credit-risk narrative. Its priors overrode the supplied evidence — under constrained prompting.

A model that will not stop

Reasoning-tuned models abstain **24% worse** than their non-reasoning counterparts (AbstentionBench). The better it reasons, the less it admits it cannot.

An industry that has named it

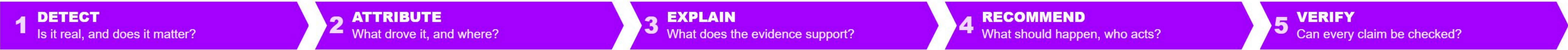
Gartner names **"agent drift"** as the risk in autonomous analytics and prescribes **guardian layers** as the mitigation.

If the default behaviour of a capable model is to answer confidently, abstention is not a feature you add later — it is an architecture you start from.

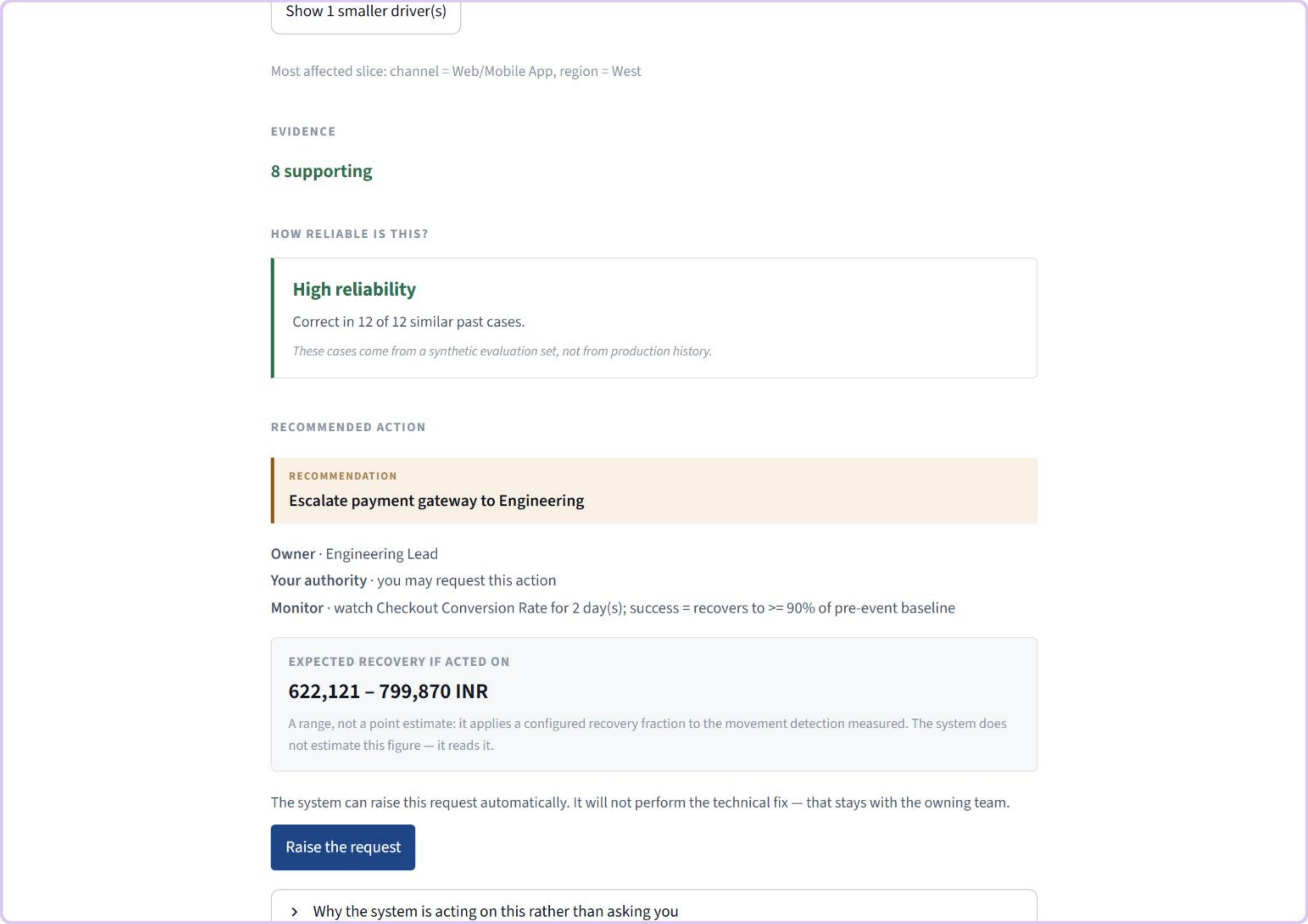
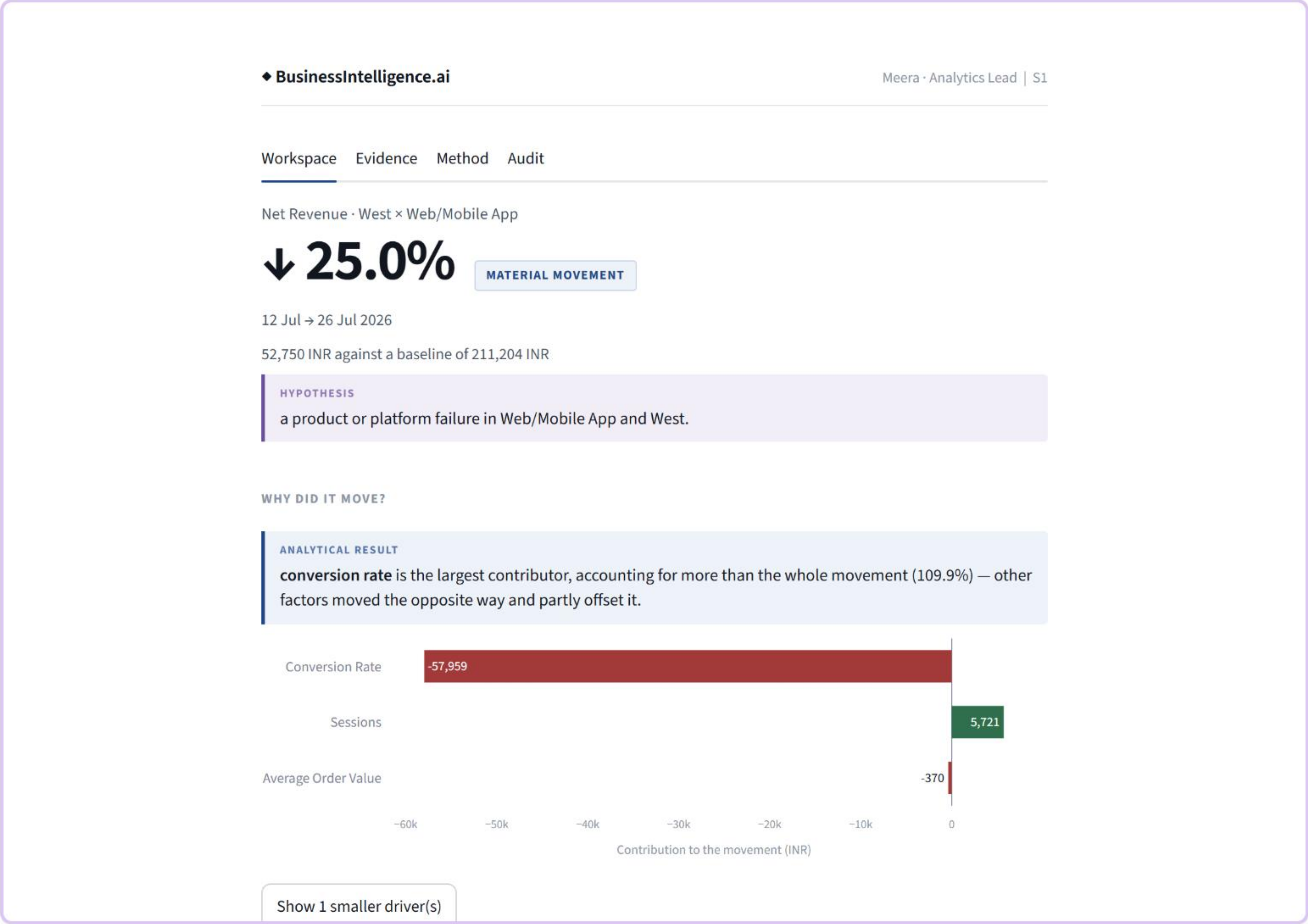
WHAT THAT MEANS FOR THE BUILD



03 Our Solution: one engine, from a number to a decision



REAL RENDERED OUTPUT — S1, WORKSPACE TAB. WHAT CHANGED → WHY → EVIDENCE → HOW RELIABLE → WHAT TO DO, IN THE ORDER A BUSINESS USER ASKS.



One screen, one decision

No chat box, by design. A conversational surface invites questions the model must not answer, from data it must not reach.

...and when the evidence conflicts, this screen does not appear at all

On S2 two explanations are equally supported and imply **different owners**. The system stops and asks a person instead of picking. **Slide 05**.

04 Round 1 promised this. Round 2 runs it, and corrects it.

Round 1 was a concept. Round 2 is 574 passing tests, 8 executable scenarios — and a list of things we said we would build and did not.

KEPT, AND NOW RUNNING

Round 1 promise	Round 2 evidence
Five stages, one engine	26-node state machine, 11 typed terminals, 8/8 scenarios end to end
Materiality gate — statistical and business	S7: a channel rename reads +5.9% growth. Real, immaterial — no alert
"Escalation is a cost rule, not a threshold"	Cost-sensitive deferral, implemented — it routes S2 to a human
The LLM narrates only what was already verified	Narration request carries no tools key — absent, not empty

CHANGED, AND WHY — INTELLECTUAL HONESTY, NOT APOLOGY

Round 1 said	Round 2 ships	Why
Confidence, zero point eight two	A calibrated band + track record	Narrative has no confidence field — nowhere to write one
CUSUM · Bayesian online detection	STL + robust MAD + PELT	The other two were not built. Claiming them is the exact failure we exist to prevent
Conformal prediction	Ranked hypotheses + cost-sensitive deferral	Conformal prediction was not implemented
Revenue dropped 8% in the West	West × Web/Mobile App, −25.0% (measured)	Quotes the movement the detector actually measured
"Two days becomes minutes"	4–50 s measured. No time-saving claim	No baseline investigation time was ever measured

Four of five changes made the claim smaller. That is what having a working system does to a pitch.

05 Every number is computed, never generated

"The model writes the sentence. It does not compute, retrieve, or decide."

Deterministic — everything the user acts on	The model — one job, inside a schema
Detection: STL + robust MAD z-score + PELT changepoint	Writes prose, into a typed schema
Materiality: a statistical leg and a business leg	Sees a frozen, hashed evidence bundle
Attribution: LMDI, residual-free	Has no database access
Localisation: Adtributor — power, surprise, succinctness	Has no tools key in the request
Causal test: difference-in-differences, parallel-trend check	Has no field to write a confidence into
Routing: pure predicates over deterministic state	Cannot choose what happens next

109.9%

Conversion rate accounts for **more than the whole movement** — sessions rose and partly offset it. An exact identity closes to **0.000000000%** residual, so the share is arithmetically correct and the system explains why rather than clipping it. **No language model generates a self-explaining share above 100%.**

Enforced by test, not prompt

test_the_client_never_offers_tools — asserted against the **recorded request**.

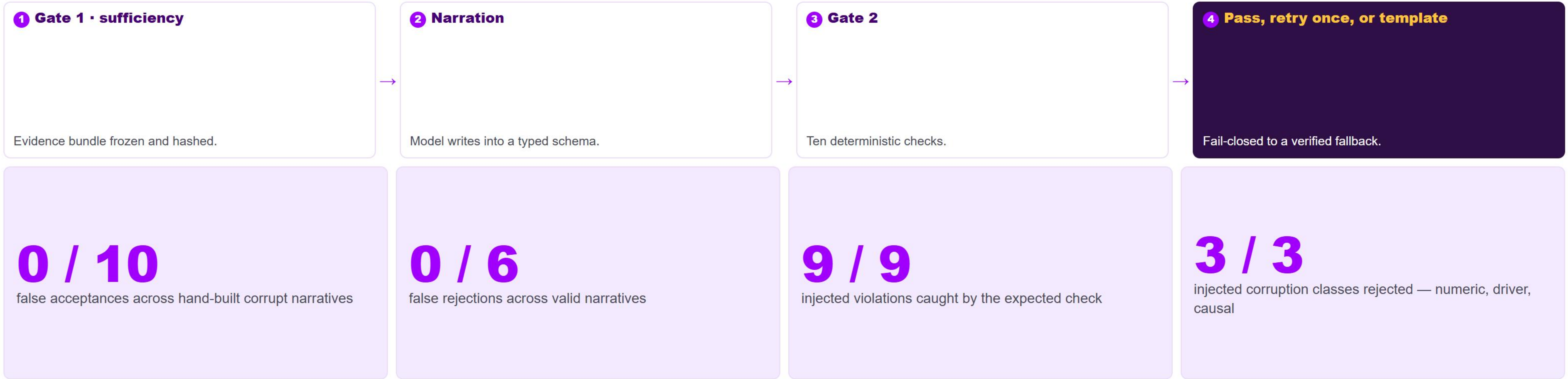
Enforced by test, not prompt

test_no_routing_predicate_reads_a_narrative — asserted against predicate **source**.

Robustness

Top driver held in **100% of 300** bootstrap resamples.

06 Nothing ships unverified — the model never gets the last word



The gate blocked our own fallback template

During the final audit, a bug made the deterministic template cite two evidence ids the frozen bundle no longer held. Gate 2 rejected it. We had documented that template as "cannot fail by construction" — it was unfailable by **luck**. Both the builder and the docstring were corrected.

A gate that only ever passes what we produce is not a gate.

Caveat, stated

None of *those* ten got through — not that none could.

And what this does not measure: the **verification mechanism** has been tested; **live LLM reliability has not**. Every figure here comes from the deterministic path and hand-built corruptions. No model generation has been observed — LIVE LLM EVALUATION PENDING.

Causal language is licensed, not asserted — a DiD test with a parallel-trend check decides. On S3 the licence is denied; the UI says "association only". DiD licenses wording, not truth.

07 It declines: abstention is the product, not a limitation

ACROSS THE 8 SYNTHETIC DEMONSTRATION SCENARIOS — A SET BUILT TO EXERCISE EVERY TERMINAL STATE. THESE ARE NOT PRODUCTION WORKLOAD RATES.

4 of 8 automated (50% of the demo set)	2 of 8 routed to a human (25% of the demo set)	2 of 8 declined (25% of the demo set)
S2 • Conflicting evidence South × Apparel, −21.9%. Two explanations equally supported — competitive pressure and stock availability — implying different owners. <i>Stops, states the question, hands it to a person.</i>	S4 • Sparse history A new category with 52 of the 56 days a seasonal baseline needs. It refuses to extrapolate. <i>Declines, and says what would change its mind.</i>	S7 • Immaterial artifact A channel rename (marketplace → Marketplace) reads as +5.9% growth. Statistically real, business-immaterial. <i>No alert, no recommendation, no invented cause.</i>
The pause is real The graph interrupts on a durable checkpoint. Resuming produces the same run with an identical evidence bundle hash.	And it limits its own claims Only HIGH has ≥10 observed cases. MEDIUM/LOW report UNCALIBRATED. On S2 the cost arithmetic to justify automating is unavailable — it could not automate that case even if it wanted to.	

A system that always answers cannot be trusted on the answer it gives.

08 Who may see what, and who may do what

M

Meera · Analytics Lead

Is this defensible enough to circulate?

Method, contribution, counterfactual, full lineage. Investigation becomes review.

P

Priya · Regional Ops Lead

Do I escalate, and to whom?

Region-scoped, named owner. **Denied crm_notes** — told how many were withheld.

A

Arjun · Finance Director

Is this material to the quarter?

Impact range, basis named. A higher configured decision value — same evidence can route differently.

The button says "Raise the request." It does not say "roll back."

Automating a request is not automating a remediation

The persona holds **request rights** on this lever, not approval rights. What gets automated is *raising an engineering request*. Executing a rollback is a different lever — no persona in this system can approve it, and it sits on a **never-automate list**. The enterprise fear is not a wrong explanation; it is a correct explanation wired to the wrong action.

32	security tests: row filters, column masks, source denials
6-stage	leak chain, with a non-vacuity control
0	restricted items reaching any stage

There is no authentication. Persona is a dropdown. Authorisation is real and tested end to end; identity is not. Enterprise IAM is V2 priority 1 — triggered by *"before any real data,"* not by a user count.

09 What we measured, and what we refuse to claim

Every number carries a class: [M] measured · [S] synthetic evaluation · [R] research-sourced · [A] assumption · [I] illustrative

Measured on this system [M]		Synthetic evaluation [S] · seed 20260821, 6 injected events	
Automated tests passing	574, 0 fail	False positives, 48 clean slices	0
Scenarios end to end	8 / 8	Injected-event recall (events built to be detectable by this method)	1.000
Graph vs direct-module agreement	8 / 8	Gate 2 — corrupted narratives blocked	10 / 10
LMDI identity closure	0.000000000%	Gate 2 — valid narratives wrongly blocked	0 / 6
Runtime per scenario	4–50 s	Ranking robustness	100% / 300
Restricted items reaching any stage	0	Retrieval — dense recall@10 (best of three)	0.778
		Calibration, HIGH band	12 of 12

The result we are proudest of is a set of numbers that got worse

A realism audit found the corpus held only 13 distinct texts across 895 records — retrieval was close to a lookup. Widened to 30 and **every score fell**: BM25 recall@10 0.957 → **0.654**, dense 0.933 → **0.778**. The lower numbers are the ones we report. What they support: **corpus realism removed the lexical shortcut, and dense retrieval became the stronger method under paraphrase** (0.778 vs 0.654 recall@10). What they do **not** support: that fusion beats dense — RRF recall@10 is **0.697**, between the two. Hybrid stays as a robustness mechanism, not as a measured win.

Searched for, and absent from the repository: customer savings · enterprise ROI · adoption numbers · market share · productivity gain · unqualified "accuracy" · "production-ready" · "real-time" · "autonomous" except as a negation

10 Prototype readiness, with the gaps named first

8 REAL — design, architecture, frontend, backend, storage, security, error handling, testing	1 PARTIAL — auth: authorisation real, authentication absent	4 LIGHTWEIGHT — hosting, CI/CD, caching, monitoring	1 DEFERRED — scaling, with named triggers
No live LLM evaluation No API key in this environment. Latency, tokens, cost unmeasured and not estimated. Harness written, unrun.	No real-data validation Every metric is synthetic. The largest open risk in the project.	No baseline measurement Without a baseline investigation time, no time-saving claim is possible.	~2 concurrent users DuckDB permits one writer; audit log writes on every read. "Scales horizontally" would be false.

Running without a model is a supported mode, not a degraded one. The UI labels it "Verified template mode." Numbers, evidence and decision are identical either way.



11 Why this wins, and what a pilot would prove

	Answers what	Answers why	Shows evidence	Says when it doesn't know	Bounded action
Dashboards	✔	✕	✕	✕	✕
Threshold alerting	✔	✕	✕	✕	✕
Manual SQL analysis	✔	✔	✔	depends on the analyst	✕
Generic BI copilots	✔	prose	✕	✕	✕
Generic RAG assistants	✕	prose	citations	✕	✕
BusinessIntelligence.ai	✔	✔	✔	✔	✔

WHERE THE VALUE COMES FROM — FOUR MECHANISMS, EACH TIED TO SOMETHING ALREADY BUILT. NO ROI FIGURE IS CLAIMED ANYWHERE IN THIS PROJECT.

Faster diagnosis

Evidence gathering and structured root-cause analysis are automated: warehouse, tickets, CRM and deploy logs reconciled in one pass.

Safer decisions

Unsupported explanations are blocked before delivery, and genuine uncertainty is distinguished from system failure rather than smoothed over.

Governed action

Recommendations are constrained to approved levers with named owners. Automation is bounded: raise a request, never execute a remediation.

Auditability

Every conclusion carries its evidence, its lineage and its decision state — 15 lineage records per run, every read audited including denials.

A good analyst does everything in that table. What they cannot do is run in fifteen seconds, apply the same materiality rule every time, or leave an audit trail that survives their absence. This product does not compete with a spreadsheet — it competes with how long the spreadsheet takes.

Now

Round 2 prototype

6 KPIs · 3 sources · 3 personas · 8 scenarios · 574 tests

V2

Enterprise pilot

replace assumptions with measurements

V3

Production platform

warehouse · workers · gateway · observability

The ask: one KPI family, one team, one quarter — measuring time-to-explanation against the current baseline. That converts our strongest argument into a data point.

A dashboard tells you what moved. An LLM will tell you why, confidently, whether or not it knows. This system tells you why, shows the evidence, recommends what to do — and tells you when it cannot.